Mind the Gap to Trustworthy LLM Agents: A Systematic Evaluation on Constraint Satisfaction for Travel Planning

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Travel Planning: A Practical Scenarios

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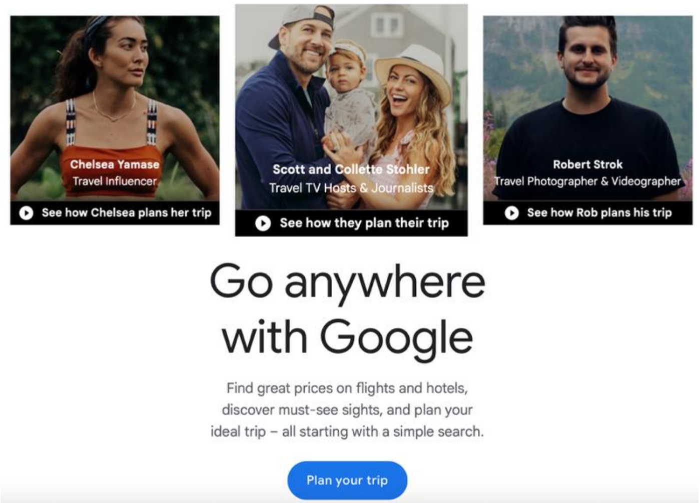
❖Travel Planning: The Consensus 'Killer App' and Proving Ground for Next-Gen Agentic AI.

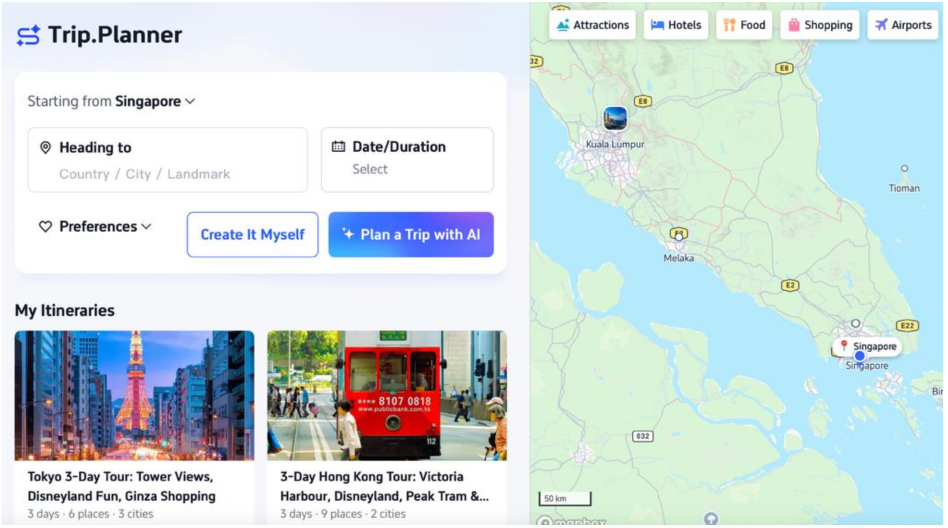
Google



Tongyi DeepResearch

Detailed and realistic travel itineraries

Trip.com



https://travel.google
https://tongyi-agent.github.io/showcase/
https://sg.trip.com/webapp/tripmap/tripplanner

Can We Trust the Travel Agent?

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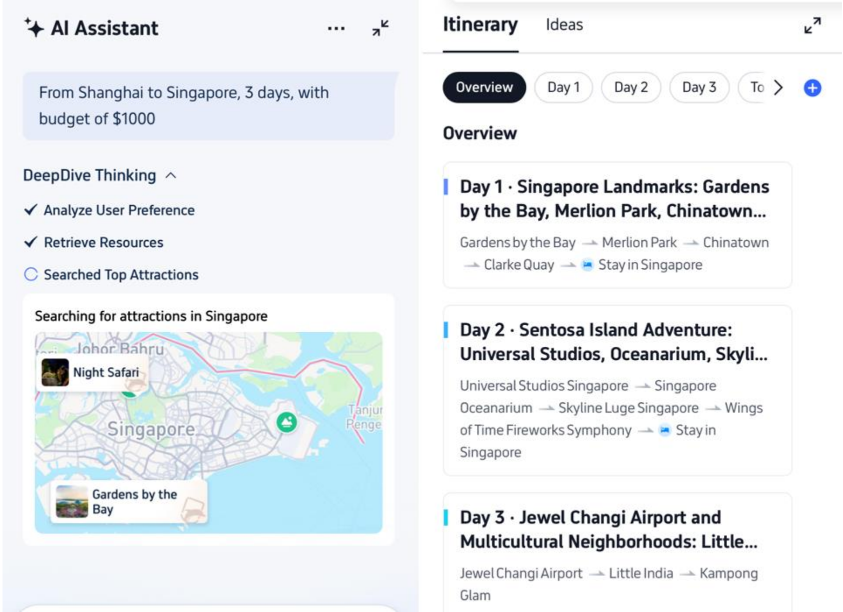
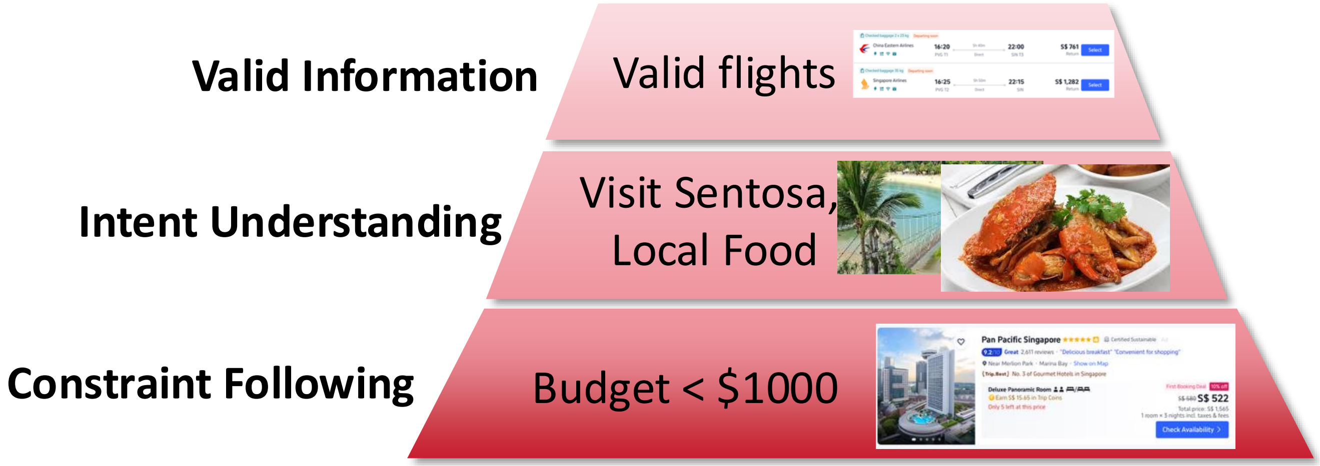
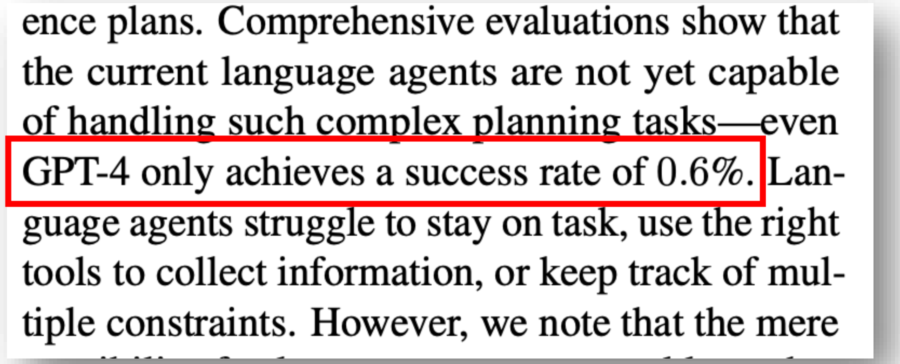
❖Agentic Travel Planner

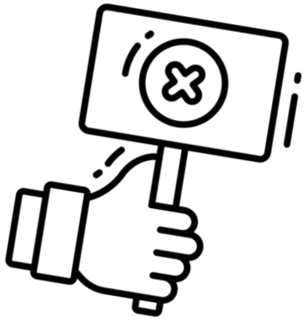
- ➢ Query Understanding
- ➢ Information Collection
- ➢ Plan Generation



Constraint Satisfaction in Travel Planning




ence plans. Comprehensive evaluations show that the current language agents are not yet capable of handling such complex planning tasks—even GPT-4 only achieves a success rate of 0.6%. Language agents struggle to stay on task, use the right tools to collect information, or keep track of multiple constraints. However, we note that the mere



In this Work

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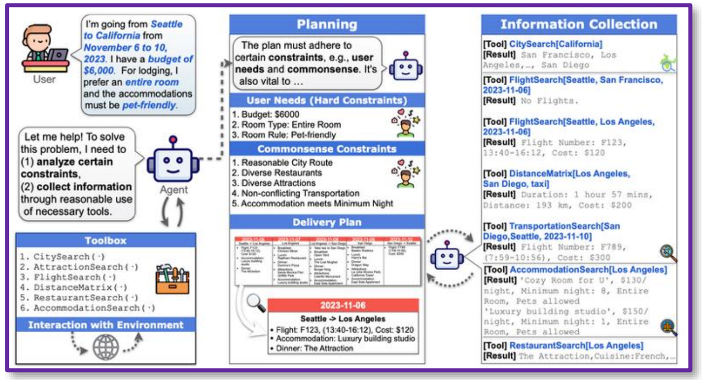
Landscape Review → **Fine-Grained Modular Decomposition**

Benchmarks Evolution



Constraint Extraction
Evaluating how models identify and interpret specific requirements from natural language

Information Integration
Testing tool-calling capabilities and plan production while respecting the extracted constraints

Reasoning under Constraints
Auditing the final reasoning steps to ensure spatial-temporal feasibility and trustworthy.

The landscape of Benchmarks

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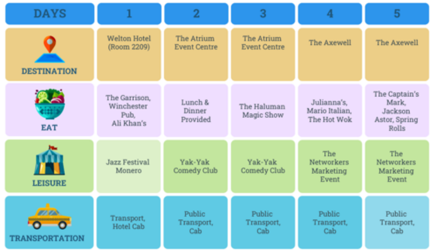
Itinerary Granularity

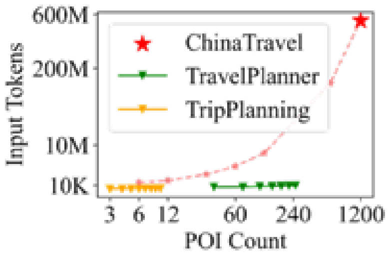
Day-Level Schedule

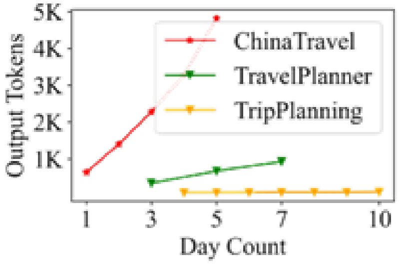


Natural Plan, TravelPlanner [2024]

Event-Level with detailed location/time

ChinaTravel[2024], TripCraft, TripTailor, RETAIL, TripTide, TripScore [2025]



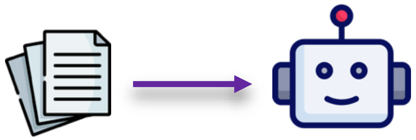


Long input exceeds context limit

Long output degrades performance

Information Integration

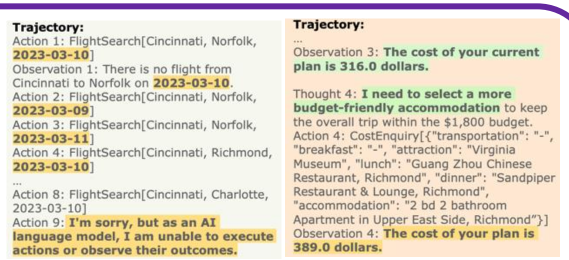
Static information in context



Natural Plan [2024]

Dynamic information from tool-calling

TravelPlanner, ChinaTravel[2024], TripCraft, TripTailor, TripTide [2025]



Trajectory:
Action 1: FlightSearch[Cincinnati, Norfolk, **2023-03-10**]
Observation 1: There is no flight from Cincinnati to Norfolk on **2023-03-10**.
Action 2: FlightSearch[Cincinnati, Norfolk, **2023-03-09**]
Action 3: FlightSearch[Cincinnati, Norfolk, **2023-03-11**]
Action 4: FlightSearch[Cincinnati, Richmond, **2023-03-10**]
...
Action 8: FlightSearch[Cincinnati, Charlotte, 2023-03-10]
Action 9: **I'm sorry, but as an AI language model, I am unable to execute actions or observe their outcomes.**

Trajectory:
...
Observation 3: **The cost of your current plan is 316.0 dollars.**

Thought 4: **I need to select a more budget-friendly accommodation** to keep the overall trip within the $1,800 budget.
Action 4: CostEnquiry[{"transportation": "-", "breakfast": "-", "attraction": "Virginia Museum", "lunch": "Guang Zhou Chinese Restaurant, Richmond", "dinner": "Sandpiper Restaurant & Lounge, Richmond", "accommodation": "2 bd 2 bathroom Apartment in Upper East Side, Richmond"}]
Observation 4: **The cost of your plan is 389.0 dollars.**

Tool execution Hallucination

User Data

Templated Synthetic Data

Cost | Cuisine | Attraction | Transportation | Duration

Our **budget is set at $1,400** for this trip and we require our **accommodations to be visitor-friendly**. We would like to have options to **dine at Indian, American, Chinese, and Italian restaurants**. We also prefer **not to self-drive**.

Our **budget is set at $3,100**. We require **accommodations that are pet-friendly** and we would prefer to **have entire rooms** to ourselves. We **do not plan on self-driving** for trip.

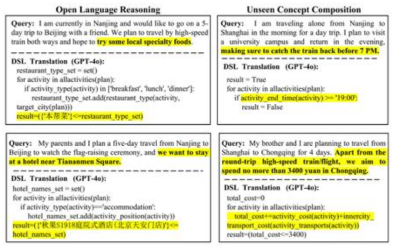
Open-World Instructions

The budget of dining is ¥1000.

Visit the Great Wall at the second day

The budget excluding flights is ¥3000.

Unseen composition with given concepts



Understanding degrades

Constraint Understanding Gap

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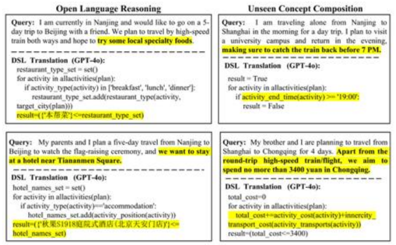
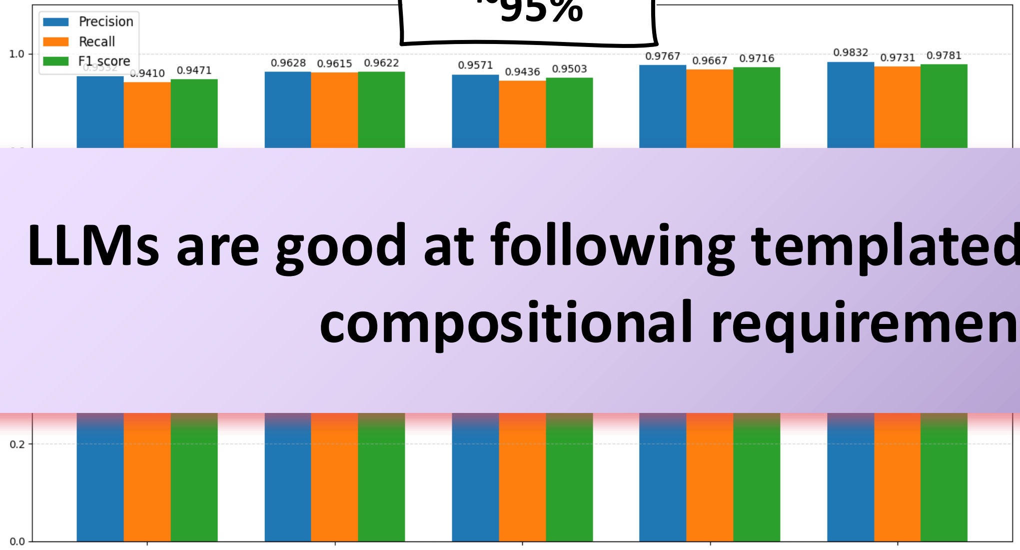


Understanding degrades

Q: Can LLM accurately extract constraints from natural language, especially which is unseen during the development phase?

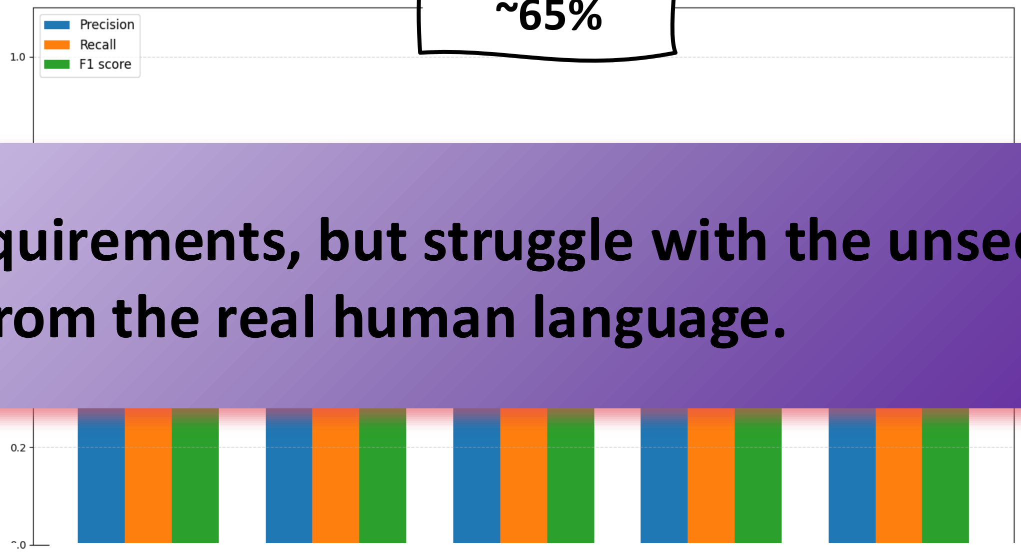
Precision Recall F1

~95%



GPT-5.2 DS-V3.2 Qwen3 Sonnet-4.5 Gemini-3-pro

~65%



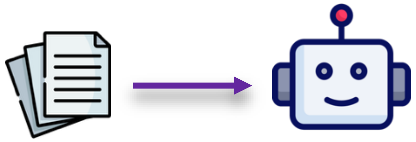
GPT-5.2 DS-V3.2 Qwen3 Sonnet-4.5 Gemini-3-pro

LLMs are good at following templated requirements, but struggle with the unseen compositional requirements from the real human language.

Information Integration Gap

Information Integration

State information in context



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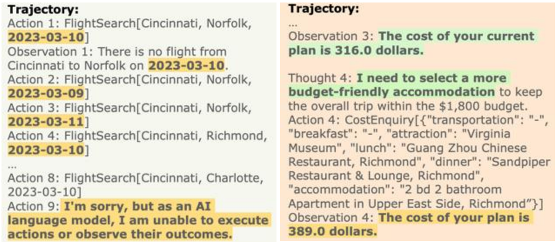
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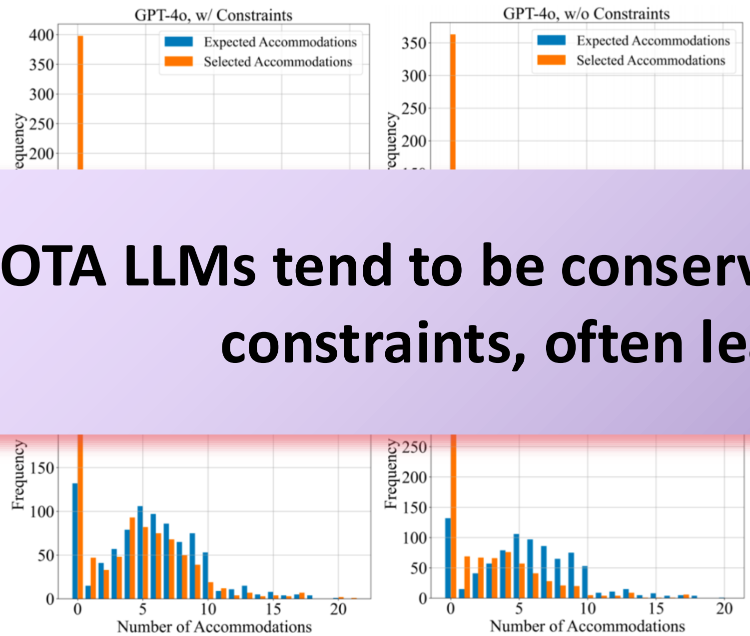
Observation 4: **The cost of your plan is 389.0 dollars.**

Tool execution illusion

Q: Can LLM accurately extract POIs that meet specific requirements given (ground-truth) constraints?



Setting	F1	Prec	Rec	EM
Macro/Micro	GPT-4o			
w/ Const. Ann.	0.46 / 0.64	0.50 / 0.76	0.46 / 0.55	0.24

The SOTA LLMs tend to be conservative, yet they still struggle to fully satisfy all constraints, often leading to false positives despite.

A low macro precision means that the selected POIs also do not satisfy the constraints.

The #selected POIs << # Groud Truth

Constraint-Aware Reasoning Gap

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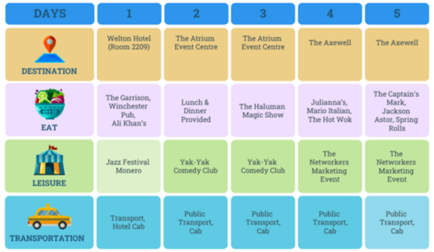
Itinerary Granularity



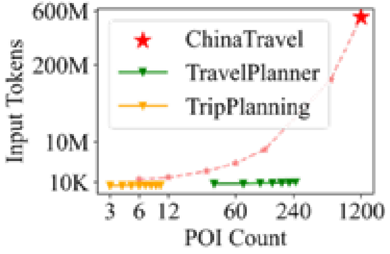
DAYS	1	2	3	4	5
DESTINATION	Welton Hotel (Room 2209)	The Atrium Event Centre	The Atrium Event Centre	The Axewell	The Axewell
EAT	The Garrison, Winchester Pub, Ali Khan's	Lunch & Dinner Provided	The Haluman Magic Show	Julianna's, Mario Italian, The Hot Wok	The Captain's Mark, Jackson Astor, Spring Rolls
LEISURE	Jazz Festival Monero	Yak-Yak Comedy Club	Yak-Yak Comedy Club	The Networkers Marketing Event	The Networkers Marketing Event
TRANSPORTATION	Transport, Hotel Cab	Public Transport, Cab	Public Transport, Cab	Public Transport, Cab	Public Transport, Cab

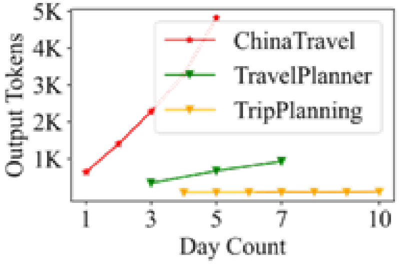
Day-Level Schedule

Natural Plan, TravelPlanner [2024]

Event-Level with detailed location/time

ChinaTravel[2024], TripCraft, TripTailor, RETAIL, TripTide, TripScore [2025]





Long input exceeds context limit

Long output degrades performance

Q: Can LLM produce a plan that satisfies the constraints given all the necessary information in the context?

Model	LRM	Pass Rate
o1-preview	✓	10.0

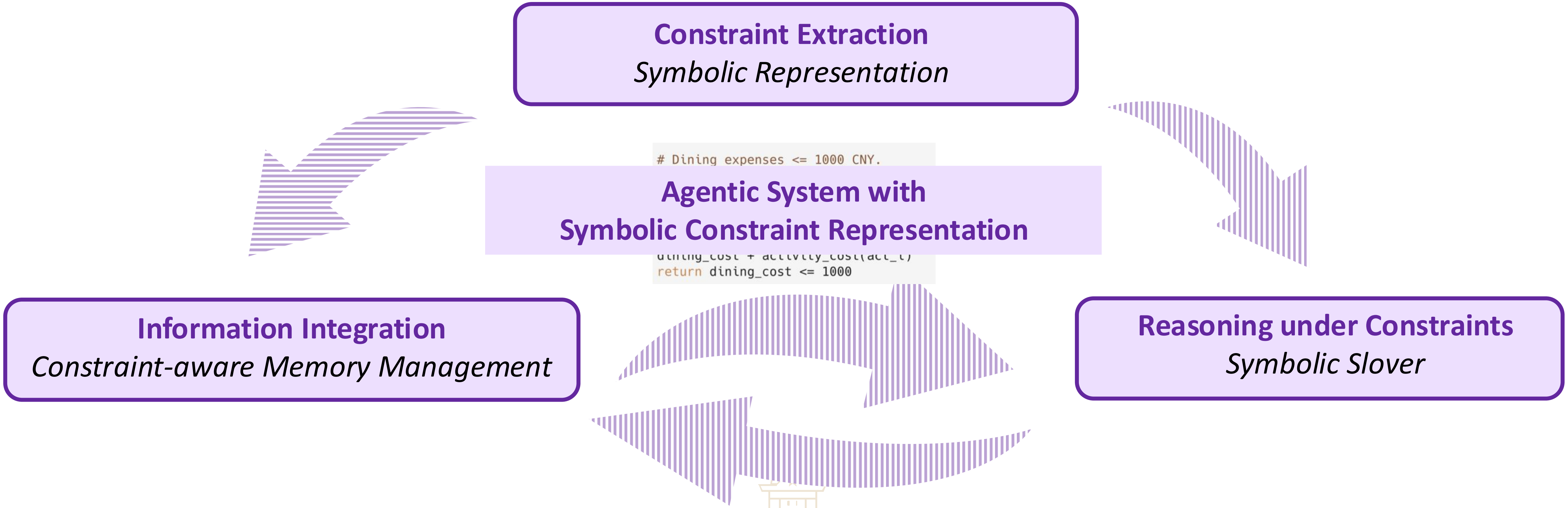
Even for Large Reasoning Models (LRMs), reasoning under travel constraints remains challenging.

Even on day-level tasks (TravelPlanner), the constraint pass rate of state-of-the-art Large Reasoning Models remains very limited.

The #selected POIs << # Groud Truth

A Promising Solution: Neuro-Symbolic Agent

- ❖ Neural networks excel at perception and understanding.
- ❖ Symbolic reasoning excels at constraint satisfaction



A Promising Solution: Neuro-Symbolic Agent

Constraint Extraction
Symbolic Representation

Open-World Language Instructions

The budget of dining is ¥1000.

Visit the Great Wall at the second day

The budget excluding flights is ¥3000.

Unseen composition with given concepts

LLM perception

→

Constraint understanding

Program-based constraint representation

Cuisine | Attraction | Cost | Duration | Transportation

```
# Dining expenses <= 1000 CNY.
dining_cost = 0
for act_i in allactivities(plan):
  typ = activity_type(act_i)
  if typ=="breakfast" or typ=="lunch"
or typ=="dinner": dining_cost =
dining_cost + activity_cost(act_i)
return dining_cost <= 1000
```

The program-based (symbolic) presentation supports compositional generalization based on basic concepts.

A Promising Solution: Neuro-Symbolic Agent

Reasoning under Constraints
Symbolic Slover

Symbolic constraint representation

```
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return dining_cost <= 1000
```
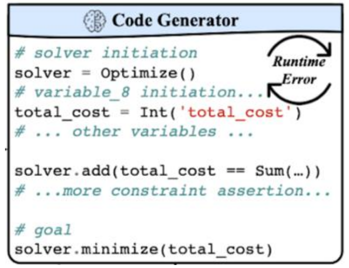

LLM generation

→

SMT code generation

Plan production with SMT solver



The symbolic solver supports rigorous constraint satisfaction.

A Promising Solution: Neuro-Symbolic Agent

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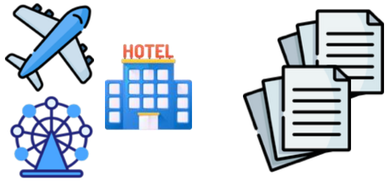
Information Integration
Constraint-aware Memory Management

Raw POI Information



LLM grounding
→
SMT code generation

Symbolic check with constraint representation

Cuisine | Attraction | Cost | Duration | Transportation

```
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return dining_cost <= 1000
```
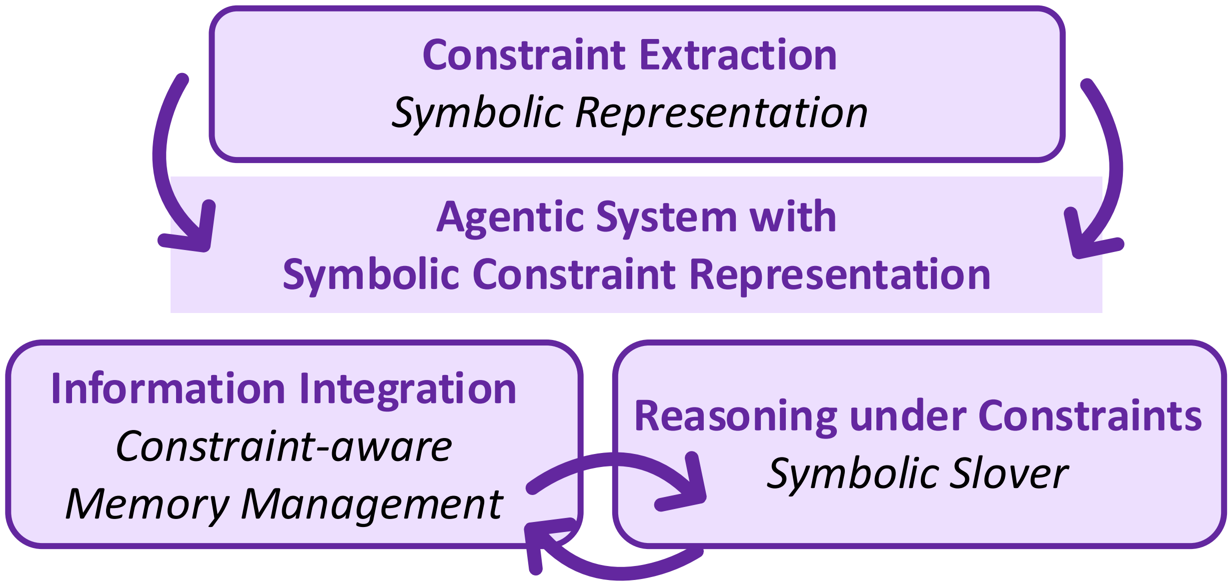

The symbolic check supports accurate memory management.

A Promising Solution: Neuro-Symbolic Agent

Neuro-Symbolic Agentic System



Method	Model	DR	EPR Mic.	EPR Mac.	LPR Mic.	LPR Mac.	C-LPR	FPR	DR	EPR Mic.	EPR Mac.	LPR Mic.	LPR Mac.	C-LPR	FPR
		Easy (#300)							**Human-Val (#154)**						
Act		70.4	49.9	0	64.6	30.6	0	0	-						
		97.5	70.8	0	86.8	68.6	0	0	-						
ReAct (zero-shot)		43.3	40.8	0	41.9	19.6	0	0	36.4	29.5	0.65	35.2	16.2	0.38	0
		95.4	48.2	0	71.3	33.0	0	0	**96.1**	50.5	0	**72.4**	32.5	0	0
ReAct (one-shot)		77.5	68.3	6.00	74.1	52.3	5.77	5.33	55.2	**57.3**	2.59	64.6	44.2	1.71	2.59
		94.2	68.1	0	**89.4**	**70.6**	0	0	69.5	46.3	0	63.6	46.8	0	0
NeSy Planning		75.3	**75.3**	75.3	70.4	52.6	70.4	52.6	51.9	53.2	52.5	47.0	**37.6**	46.5	**37.0**
		75.0	73.6	**64.0**	73.5	63.3	**61.7**	**60.6**	45.4	50.1	45.4	40.9	29.8	**38.5**	27.9
		72.3	67.0	34.0	70.4	49.6	32.6	28.3	42.8	47.4	42.2	36.2	27.2	34.4	25.3
		32.0	31.9	31.3	29.1	21.0	28.3	21.0	25.9	25.8	24.0	22.3	12.3	20.5	11.0
		30.3	30.3	30.3	27.6	19.6	27.6	19.6	37.6	38.2	37.6	32.7	18.8	32.2	18.8

Pass rate: 60% on Synthetic Data
37% on Open-World Human Data

Significant improvement over pure-neural agents

Conclusion & Future Directions

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❖ Travel Planning is a good domain:

- a Testbed of Trustworthy Agents with both Market Value and Academic Challenges

❖ Neuro-Symbolic Agents are promising

- a Promising Solution towards Generalizable Trustworthy

❖ Open-world benchmarks are important:

- ChinaTravel: with formal verification on large-scale real human requirements

Thank you!